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ASSIGNMENT 4

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HPS, Chapter 1: Exercises 1, 3, 4, 5, 6, 7

Problems from the Exercises at the end of the first chapter: 1, 3, 4, 5, 6, 7. The exercises are HPS's, pp. 10–12, reproduced verbatim, and the solutions keep the book's symbols: $\mathcal{L}(\theta, a)$ for the loss, $\mathcal{R}(\theta, d) = E^\theta\{\mathcal{L}(\theta, d(X))\}$ for the risk function (Def. 1, p. 6), and $r(\pi, d) = E\{\mathcal{R}(\theta, d)\}$ for the mean risk under a prior π (Def. 3, p. 9). The book writes the loss with θ first where the lecture writes $L(a, \theta)$; for the symmetric losses here nothing depends on the order. Every risk below is assembled by the bias–variance decomposition (notes, Supplement 1.A.1).

Exercise 1

Given $f(x | \theta) = \frac{e^{-(1/2)(x-\theta)^2}}{\sqrt{2\pi}}$ and $\mathcal{L}(\theta, a) = (a - \theta)^2$, suppose a single observed value of X is to be taken. Calculate the value of $\mathcal{R}(\theta, d)$ if d is chosen to be the function $d(x) = cx$, where c is a constant.

SOLUTION:

The model is one observation $X \sim N(\theta, 1)$, so $E^\theta\{X\} = \theta$ and $\text{Var}^\theta(X) = 1$.

$$E^\theta\{cX\} = c\theta \quad (\text{linearity})$$

$$b(\theta) = c\theta - \theta = (c - 1)\theta \quad (\text{definition of bias})$$

$$\text{Var}^\theta(cX) = c^2 \text{Var}^\theta(X) = c^2 \quad (\text{scaling; unit variance})$$

$$\mathcal{R}(\theta, d) = \boxed{c^2 + (c - 1)^2 \theta^2} \quad (\text{bias-variance decomposition})$$

The anatomy is worth reading before Exercise 4 prices it. The variance contributes a flat floor c^2 ; the bias contributes a parabola in θ that vanishes only at $c = 1$, whose rule $d(x) = x$ has constant risk 1. Any $c < 1$ shrinks the observation toward 0, trading a lower floor for a risk that grows with θ^2 : the normal cousin of the shrinkage families of Assignments 2 and 3, and exactly the class $\{d(x) = cx\}$ that the book runs through its worked Poisson example (HPS, pp. 8–10). ■

Exercise 3

Given $f(x | \theta) = \binom{2}{x} \theta^x (1 - \theta)^{2-x}$, $x = 0, 1, 2$, $0 < \theta < 1$, and $\mathcal{L}(\theta, a) = (a - \theta)^2$,

- (a) calculate $\mathcal{R}(\theta, d)$ for $d(x) = x/2$;
- (b) calculate $\mathcal{R}(\theta, d)$ for $d(x) = (x + 1)/4$;
- (c) Which of these two functions is the minimax function with respect to the two functions?
- (d) Is one decision function superior to the other?

SOLUTION:

Here $X \sim \text{Binomial}(2, \theta)$, so $E^\theta\{X\} = 2\theta$ and $\text{Var}^\theta(X) = 2\theta(1 - \theta)$.

- (a) The observed rate is unbiased, and its risk is its variance:

$$E^\theta\{X/2\} = \frac{2\theta}{2} = \theta \quad (\text{linearity; unbiased})$$

$$\mathcal{R}(\theta, d_1) = \text{Var}^\theta(X/2) = \frac{2\theta(1-\theta)}{4} = \boxed{\frac{\theta(1-\theta)}{2}} \quad (\text{scaling; bias-variance})$$

- (b) The second rule is an old friend in disguise: $d_2(x) = (x + 1)/4 = (1 + x)/(2 + n)$ at $n = 2$ —the lecture’s shaded estimator, the Bayes rule for the uniform prior, Laplace’s rule of succession (notes, Example 1.8.3 and Supplement 1.A.5).

$$E^\theta\{d_2(X)\} = \frac{2\theta + 1}{4}, \quad b(\theta) = \frac{2\theta + 1}{4} - \theta = \frac{1 - 2\theta}{4} \quad (\text{linearity; algebra})$$

$$\text{Var}^\theta(d_2(X)) = \frac{2\theta(1-\theta)}{16} = \frac{\theta(1-\theta)}{8} \quad (\text{shift and scale})$$

$$\mathcal{R}(\theta, d_2) = \frac{\theta(1-\theta)}{8} + \frac{(1-2\theta)^2}{16} = \boxed{\frac{2\theta^2 - 2\theta + 1}{16}} \quad (\text{bias-variance; simplify})$$

the simplification being $2\theta(1 - \theta) + (1 - 2\theta)^2 = 2\theta - 2\theta^2 + 1 - 4\theta + 4\theta^2 = 2\theta^2 - 2\theta + 1$.

(c) Compare the worst cases (HPS, Def. 2, p. 7). The first risk is largest in the middle,

$$\sup_{0 < \theta < 1} \mathcal{R}(\theta, d_1) = \mathcal{R}\left(\frac{1}{2}, d_1\right) = \frac{1}{8},$$

while the second, equal to $(1 - 2\theta(1 - \theta))/16$, is largest at the *edges*, approaching

$$\sup_{0 < \theta < 1} \mathcal{R}(\theta, d_2) = \frac{1}{16} \quad (\theta \rightarrow 0 \text{ or } 1).$$

Since $\frac{1}{16} < \frac{1}{8}$,

$$d_2(x) = (x + 1)/4 \text{ is minimax with respect to the pair}$$

its worst case being half of d_1 's.

(d) Neither is superior. Setting $\mathcal{R}(\theta, d_1) \leq \mathcal{R}(\theta, d_2)$ and writing $u = \theta(1 - \theta)$, the comparison $8u \leq 1 - 2u$ reduces to $u \leq \frac{1}{10}$, i.e.

$$\theta \leq \frac{5 - \sqrt{15}}{10} \approx 0.113 \quad \text{or} \quad \theta \geq \frac{5 + \sqrt{15}}{10} \approx 0.887,$$

so d_1 wins near the edges and d_2 wins on the wide middle (Figure 1): the risk functions cross, and—in the book's own words for exactly this situation—“neither function is superior to the other” (HPS, p. 6).

Neither rule is superior: each has smaller risk on part of $(0, 1)$

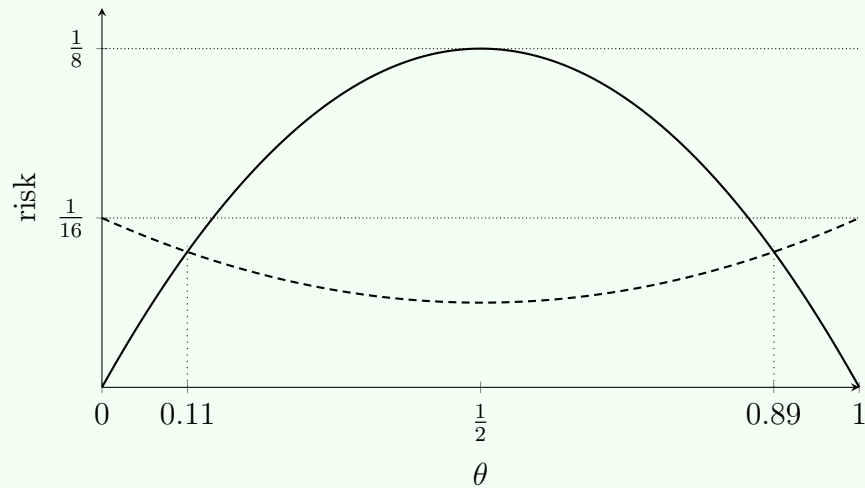


Figure 1. THE TWO RISKS OF EXERCISE 3. The observed rate $d_1 = x/2$ (solid) peaks at $\frac{1}{8}$ in the middle; the shaded rule $d_2 = (x + 1)/4$ (dashed) is worst at the edges, approaching $\frac{1}{16}$. The dotted levels are the two suprema— d_2 's worst case is half of d_1 's, so d_2 is minimax within the pair—and the vertical dotted lines mark the crossings at $\theta = (5 \mp \sqrt{15})/10$: each rule wins somewhere, so neither is superior.

■

Exercise 4

Given the prior density $\pi(\theta) = 1/2$, $-1 < \theta < 1$;

- (a) calculate the mean risk for Exercise 1;
- (b) find the value of c which will produce a Bayes solution with respect to this prior distribution.

SOLUTION:

(a) The mean risk integrates Exercise 1's risk against the uniform prior on $(-1, 1)$, whose only relevant moment is $E\{\theta^2\} = \int_{-1}^1 \theta^2 \cdot \frac{1}{2} d\theta = \frac{1}{3}$ (the prior mean is 0 by symmetry):

$$r(\pi, d) = E\{c^2 + (c - 1)^2\theta^2\} = \boxed{c^2 + \frac{(c - 1)^2}{3}}$$

(integrate the risk; prior second moment)

(b) Minimize over c by completing the square:

$$\begin{aligned} r(\pi, d) &= \frac{4}{3}c^2 - \frac{2}{3}c + \frac{1}{3} && \text{(expand)} \\ &= \frac{4}{3}\left(c - \frac{1}{4}\right)^2 + \frac{1}{4} && \text{(complete the square)} \end{aligned}$$

so the mean risk is minimized at

$$\boxed{c = \frac{1}{4}} \quad \text{with} \quad r(\pi, d) = \frac{1}{4} :$$

within the class $\{d(x) = cx\}$, the Bayes solution for this prior is $d(x) = x/4$. The number has the familiar anatomy: with observation variance 1 and prior scatter $v = E\{\theta^2\} = \frac{1}{3}$, the minimizing coefficient is $c^* = v/(1 + v)$ —the prior's share of the total

spread, shrinking the observation three-quarters of the way toward the prior mean 0. It is the same precision arithmetic as Assignment 3's $a^* = \sigma^2/(n + \sigma^2)$, and the computation mirrors the book's own within-class Bayes calculation for the Poisson, where $\pi(\theta) = e^{-\theta}$ produced $c = \frac{1}{2}$ (HPS, p. 10). A wide prior (v large) would push c^* toward the unbiased $c = 1$; this narrow one, confining θ to $(-1, 1)$, buys a mean risk of $\frac{1}{4}$ against the constant risk 1 of $d(x) = x$. ■

Exercise 5

Given $f(x \mid \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}$, $x = 0, 1, \dots, n$, $0 < \theta < 1$, and $\mathcal{L}(\theta, a) = (a - \theta)^2$,

- (a) calculate $\mathcal{R}(\theta, d)$ for $d(x) = x/n$;
- (b) given $\pi(\theta) = 1$, $0 < \theta < 1$, calculate $r(\pi, d)$ for $d(x) = x/n$.

SOLUTION:

- (a) With $X \sim \text{Binomial}(n, \theta)$, the rate is unbiased and its risk is its variance:

$$E^\theta\{X/n\} = \frac{n\theta}{n} = \theta \quad (\text{linearity; unbiased})$$

$$\mathcal{R}(\theta, d) = \text{Var}^\theta(X/n) = \frac{n\theta(1-\theta)}{n^2} = \boxed{\frac{\theta(1-\theta)}{n}} \quad (\text{scaling; bias-variance})$$

- (b) Under the uniform prior, using $\int_0^1 \theta(1-\theta) d\theta = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$,

$$r(\pi, d) = \int_0^1 \frac{\theta(1-\theta)}{n} d\theta = \boxed{\frac{1}{6n}} \quad (\text{integrate the risk})$$

This pair of numbers is the course's running baseline: the risk is the R_{d_0} of Assignment 2 and the per-arm risk of Assignment 3's trial, and the mean risk $1/(6n)$ is the quantity the shaded estimator improves to $1/(6(n+2))$ (notes, Supplements 1.A.3 and 1.A.5). ■

Exercise 6

Work Exercise 5 if $\mathcal{L}(\theta, a) = (a - \theta)^2/\theta(1 - \theta)$.

SOLUTION:

The weight $1/\theta(1 - \theta)$ is a function of θ alone, so at each fixed θ it passes straight through the expectation over the data:

$$\mathcal{R}(\theta, d) = \frac{\mathbb{E}^\theta\{(X/n - \theta)^2\}}{\theta(1 - \theta)} = \frac{\theta(1 - \theta)/n}{\theta(1 - \theta)} = \boxed{\frac{1}{n}}$$

(weight is constant in x ; Exercise 5(a))

a *constant* risk; and the mean risk of a constant is itself,

$$r(\pi, d) = \int_0^1 \frac{1}{n} d\theta = \boxed{\frac{1}{n}} \quad (\text{expectation of a constant})$$

The exercise is quietly making a point, and it is worth making out loud. This is the loss the book licenses in §1.4 with the family $\mathcal{L}(\theta, a) = c(\theta)|a - \theta|^b$ (p. 10), and the weight is chosen so that errors count relative to the difficulty of the problem at θ —misses near the edges, where the variance is tiny, are charged heavily. Under that accounting the observed rate becomes an *equalizer*: its risk curve, a bump under plain squared error, flattens to the line $1/n$. And the flattening is no accident: for a loss of the form $w(\theta)(a - \theta)^2$, the action minimizing the posterior expected loss is the reweighted posterior mean $\mathbb{E}\{w(\theta)\theta \mid x\}/\mathbb{E}\{w(\theta) \mid x\}$, and with the uniform prior the reweighting turns the $\text{Beta}(x+1, n-x+1)$ posterior kernel into $\theta^{x-1}(1-\theta)^{n-x-1}$ —the $\text{Beta}(x, n-x)$ kernel—whose mean is exactly x/n (for $1 \leq x \leq n-1$; at $x=0$ or n the weighted integrals diverge and the natural boundary action agrees with x/n). So under this loss the rate is the Bayes rule for the uniform prior, and a Bayes rule with

constant risk is minimax—the weighting rehabilitates x/n from Exercise 3’s loser into the optimal rule. ■

Exercise 7

Given $f(x \mid \theta) = \frac{e^{-\theta} \theta^x}{x!}$, $x = 0, 1, \dots$, $\theta > 0$, and $\mathcal{L}(\theta, a) = (a - \theta)^2$,

(a) calculate $\mathcal{R}(\theta, d)$ for $d(x) = x$.

(b) Given the gamma density $\pi(\theta) = \frac{\lambda^\alpha \theta^{\alpha-1} e^{-\lambda\theta}}{\Gamma(\alpha)}$, $\alpha > 0$, $\lambda > 0$, $\theta > 0$, calculate $r(\pi, d)$ for $d(x) = x$.

SOLUTION:

(a) The Poisson has mean and variance both equal to θ : the mean is $E^\theta\{X\} = \sum_{x \geq 1} x e^{-\theta} \theta^x / x! = \theta \sum_{x \geq 1} e^{-\theta} \theta^{x-1} / (x-1)! = \theta$, and the factorial-moment trick $E^\theta\{X(X-1)\} = \theta^2$ (shift the summation index twice) gives $E^\theta\{X^2\} = \theta^2 + \theta$, hence $\text{Var}^\theta(X) = \theta$. The rule $d(x) = x$ is therefore unbiased, and

$$\mathcal{R}(\theta, d) = \text{Var}^\theta(X) = \boxed{\theta} \quad (\text{unbiased; bias-variance})$$

Unlike every risk so far, this one is unbounded: on $\Theta = (0, \infty)$ the worst case is infinite, so minimax reasoning has nothing to grip—which is precisely why the book pairs this model with a prior.

(b) The mean risk is the prior expectation of θ , and the gamma integral is one line of Γ -bookkeeping:

$$r(\pi, d) = \int_0^\infty \theta \frac{\lambda^\alpha \theta^{\alpha-1} e^{-\lambda\theta}}{\Gamma(\alpha)} d\theta = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} = \boxed{\frac{\alpha}{\lambda}} \quad (\Gamma(\alpha+1) = \alpha \Gamma(\alpha))$$

the gamma prior's mean: an unbounded risk curve, tamed to a finite number by a prior that puts little weight where the risk is large. The book's worked example runs the same Poisson model with the θ -weighted loss and the class $\{cx\}$ (HPS, pp. 8–10);

its prior $\pi(\theta) = e^{-\theta}$ is the case $\alpha = \lambda = 1$ here, for which $r(\pi, d) = 1$ —consistent, since under that exponential prior the unweighted risk θ averages to the prior mean 1. ■

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